

JOSE ALBERTO TUPAYACHI SILVA

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Summary

Industrial & Systems Engineer specializing in AI and Data Science, with strong presentation, problem-solving, and teamwork skills. Experienced in developing end-to-end AI solutions, working in Agile sprints and communicating technical concepts.

Education

University of Tennessee, Knoxville

PhD., Industrial Engineering May 2024 - May 2027

M.S., Industrial Engineering Aug 2022 - May 2024

Experience

Oak Ridge National Laboratory

Mar 2025 - Present

Developer

Oak Ridge, TN

- Deploy a pipeline for spectral (PCA - cuML) and spatial (alpha-shape) ToF-SIMS - 6PPD Analysis based on .itax/itmX IONTOF .
- Fine-tune a vision (Swin Transformer) to quality screen wind speed and direction from weather stations used in atmospheric plume modeling.
- Develop an explainable ML model for atmospheric and geospatial datasets (DL and 1DCNN).
- Automate geospatial dashboard generator with natural language knowledge graph reasoning (llama.cpp, langchain, vLLM).
- Deploy retrieval augmented generation for 1000+ lithium breeder blanket papers with research direction identification (ontology).

University of Tennessee, Knoxville

Aug 2022 - Present

Graduate Research Assistant

Knoxville, TN

- Deployed an AI-powered digital twin for freight transportation with linear optimization, real-time routing, and SLURM parallel processing.
- Coded a mobile health applications with Android, iOS, watchOS, and Wear OS distributions (Flutter).
- Integrated an Intermodal routing optimizer with ontology-guided REST API integration.
- Developed a Traffic-Weather Graph Convolutional Network for EV charging demand forecasting that feeds Traffic Message Channels into a ensemble architecture.

Indra

Mar 2022 - Aug 2022

Data Engineer

- Built and maintained scalable ETL/ELT pipelines using Python, Scala, and Shell scripting on UNIX/Linux, processing high-volume datasets across Agile delivery cycles to support downstream analytics and reporting.
- Engineered distributed data processing systems using Scala/PySpark on Apache Spark, Hadoop, and Hive/HQL, and led migration of a legacy Oracle warehouse to a partitioned AWS-based cloud lakehouse (S3, EMR), optimizing Spark memory and resource allocation to improve job execution efficiency.
- Designed data models to support data warehousing, and implemented entity-resolution and deduplication logic to unify and standardize records across varied sources.
- Automated CI/CD deployment pipelines using Jenkins, streamlining builds, unit testing, and application deployment including live deployments, rollback procedures, and backup strategies to reduce manual effort and deployment risk.

Globokas

Jan 2022 - Mar 2022

Business Intelligence Analyst

- Developed dashboards and KPI reporting systems supporting strategic and operational decision-making for multiple business clients.
- Integrated heterogeneous data sources (REST APIs, internal systems, external platforms) into unified analytics pipelines.

Data Analyst

- Built interactive Power BI and Tableau dashboards for operational monitoring and KPI reporting.
- Managed SQL/T-SQL databases and Salesforce integrations via API queries.
- Developed a PyQt desktop application to automate invoice verification and special-case processing.
- Engineered customer segmentation models using unsupervised machine learning (K-Means and hierarchical clustering) on payment history and demographics, enabling collection teams to prioritize outreach and improve payment-recovery.

GPS Management

Aug 2020 - Nov 2020

Developer - Portfolio Valuation

- Implemented machine learning models for non-performing loan (NPL) recovery and portfolio valuation.
- Automated scoring workflows, reducing manual processing time.

Quiron Salud

Jan 2019 - Aug 2019

IT Intern

- Developed and maintained healthcare plans information systems.
- Supported electronic health record integration and database administration for clinical operations.

Selected Publications

- Tupayachi, J., Khan, A. N., & Li, X. (2026). Scalable decentralized prognostics for industrial systems under data heterogeneity. *Computers & Electrical Engineering*, 133, Article 111023. <https://doi.org/10.1016/j.compeleceng.2026.111023>
- Khan, A. N., Tupayachi, J., Dabbs, W., Chamberlin, S., Yao, B., & Li, X. (2026). CliniSense AI for automated clinical skills assessment with real-time feedback. *npj Digital Medicine*. <https://doi.org/10.1038/s41746-026-03100-8>
- Wyatt, T. H., Lowe, S., Tupayachi, J., & Li, X. (2026). Design and usability testing of SmartSHOTS: A mobile app to reduce vaccine barriers for children 0–24 months. *Health Informatics Journal*, 32(1). <https://doi.org/10.1177/14604582261419334>
- Tupayachi, J., Ferguson, M. M., & Li, X. (2024). A simulation-based real-time deep reinforcement learning approach for fighting wildfires. *Proceedings of the 2024 Annual Modeling and Simulation Conference (ANNSIM)*, 1–12. <https://doi.org/10.23919/annsim61499.2024.10732085>
- Xu, H., Li, X., Tupayachi, J., Lian, J. J., & Omitaomu, O. A. (2024). Automating bibliometric analysis with sentence transformers and RAG. *Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Advances in Urban-AI*, 1–10. <https://doi.org/10.1145/3681780.3697252>
- Li, X., Tupayachi, J., Sharmin, A., & Martinez Ferguson, M. (2023). Drone-aided delivery methods, challenges, and the future: A methodological review. *Drones*, 7(3), Article 191. <https://doi.org/10.3390/drones7030191>
- Tupayachi, J., Xu, H., Omitaomu, O. A., Camur, M. C., Sharmin, A., & Li, X. (2024). Towards next-generation urban decision support systems through AI-powered construction of scientific ontology using large language models. *Smart Cities*, 7(5), 2392–2421. <https://doi.org/10.3390/smartcities7050094>
- + Publications: <https://scholar.google.com/citations?user=Kyn6Et8AAAAJ&hl=en&oi=ao>

Skills

AI and Machine Learning

- Machine Learning Engineering, LLMs, RAG, Agentic AI Workflows, CNNs, Vision Transformers, TensorFlow, PyTorch, Statistical Modeling, Predictive Modeling, Clustering, Regression, Tree-Based and Boosting Models (Random Forest, LightGBM, XGBoost), Model Tuning and Calibration, Explainable AI

LLM and Agentic Frameworks

- LangChain, LangGraph, LiteLLM, vLLM, Ollama, QLoRA Finetuning

Cloud and Infrastructure

- AWS (S3, EMR, Redshift, SageMaker, ECS), Microsoft Azure, Google Cloud, Docker, Kubernetes, UNIX/Linux, KVM, Slurm, CI/CD, Jenkins, Git, GitLab

Data Engineering

- Python, SQL, Scala, Apache Spark, Hadoop/MapReduce, Hive, Kafka, ETL/ELT, Batch and Real-Time Streaming, Delta Lake, Databricks Workflows, Unity Catalog, dbt, Airflow, Prefect, Data Modeling, Entity Resolution, Data Quality, Anomaly Detection, Snowflake, Amazon Redshift, Dimensional Modeling, Data Warehouse Architecture, Query and Performance Optimization

Databases

- PostgreSQL, MySQL, SQL Server, MongoDB, Cassandra

Programming Languages

- Java, R, SAS, STATA, Bash, Dart, PHP, C++, CUDA, SPSS

Web, Frameworks, and APIs

- Django, Flask, Streamlit, Dash, React, Flutter, Laravel, REST and GraphQL APIs

Optimization and Simulation

- Gurobi, CPLEX, AnyLogic, NetworkX

Geospatial

- WMS, GeoServer, OverPass API, Turf.js

Certifications

- MS Transact-SQL (T-SQL), Microsoft Fabric DP-600 & DP-700, AWS, GCP

Awards & Honors

- IISE Future Faculty Fellows (2025-2026), USA - Awarded
- Best Paper Award, CIE51 (51st Conference on Computers and Industrial Engineering), Sydney, Australia (2024)
- IISE DAIS Student Mobile App Competition - Winner, SmartShots Project, Montreal (2024)
- HIDA Helmholtz Visiting Researcher, Karlsruhe Institute of Technology (KIT), Germany - Awarded (2024)
- Holiday Fellowship, University of Tennessee, Knoxville (2022, 2023, 2024)

Clients & Sponsors

ORNL	github.com/jtupayachi/smartgiscavp	demo2.recoil.ise.utk.edu	Mar 2025 - Present
<i>Developer</i>			
• Funding: Nuclear Safety Research and Development (NSRD).			
RECOIL	ml.recoil.ise.utk.edu		Jul 2024 - Mar 2026
<i>Lead AI/Data Engineer</i>			
• Funding: U.S. Department of Energy ARPA-E, Project #DE-AR0001780.			
SmartShots	iise.org/Details.aspx?id=33697	play.google.com/store/apps/details?id=com.ilab.smartshots	Dec 2023 - Dec 2024
<i>Full-Stack Developer</i>			
• Winner, IISE DAIS Student Mobile App Competition 2024.Funding: Tennessee Department of Health.			
Active Caregiver's Toolkit	play.google.com/store/apps/details?id=com.ilabutk.fe_rich		Jul 2024 - Present
<i>Mobile Developer</i>			
• Funding: Hillman Emergent Innovation (HEI).			
ORNL	github.com/jtupayachi/tof-sims-driver	github.com/jtupayachi/Jose_LLM_OCR_PDF_Parser/tree/main	Mar 2025 - Present
<i>Developer</i>			
• Funding: Fusion Energy Sciences (FES)			